

Air pollutant dispersion around high-rise buildings due to roof emissions

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ARTICLE INFO

Keywords:

Pollutant dispersion
High-rise building
Wind-structure interaction
Roof discharge location
Computational fluid dynamics

ABSTRACT

Air quality in the built environment has a significant impact on public health, particularly in densely populated cities where residents live in closely-packed high-rise buildings. In this case, communicable respiratory diseases are apt to spread through vent pipe emissions on the building roof and facilitate community transmission. This study applied computational fluid dynamics to simulate the dispersion of the pollutants discharged from the roof vent pipes around high-rise buildings. The results show that the dispersion process is dominated by wind-structure interaction, wind incidence angle, air pollutant discharge location, and vent pipe height, and the dispersion is more complex around crucifix-form buildings than cubic ones. When the wind incidence angle reaches around 90°, the air pollutants are entrained in the re-entry of the building, thus impeding the dispersion process. The building areas most adversely affected by air pollutants are located on high floors near the vent pipe. Changing air pollutant discharge location and increasing the height of the vent pipe can significantly reduce the affected area. Where the height of the vent pipe is restricted, parapets at the roof edges are suggested to further reduce the adverse effects of air pollutants. These research findings can help to reduce the risk of airborne contaminants/diseases spreading in current and future built environments.

1. Introduction

1.1. Background

Over the past four decades, many researchers have studied the dispersion of air pollutants (e.g. Refs. [1–3], with recent interest focusing on their effect on human health. The spread of communicable diseases in the built environment remains a constant threat to human life and has raised our awareness of the impact of high-rise buildings on the dispersion of airborne contaminants. The deadly severe acute respiratory syndrome (SARS) outbreak in the residential estate of Amoy Garden in Hong Kong in 2003 highlighted the role played by wind-structure interaction in potentially facilitating the spread of airborne diseases. The spatial distribution of the infected patients was positively correlated with the spatial distribution of the virus aerosol released from the index patient's room and driven by the airflow generated by wind-structure interaction [4,5].

Like SARS, the COVID-19 pandemic has created chaos worldwide; recent studies have indicated that COVID-19 is also mainly transmitted by droplets produced via coughing or sneezing. However, viruses that

become suspended in air can be transmitted over long distances by wind-driven airflow [6,7]. A recent study has suggested that the virus is still infectious when diluted 10 000 times in the air [6]. Further evidence indicates that COVID-19 can spread via airflow without face-to-face contact [8]. Given that COVID-19 has an incubation period of approximately two weeks [9], the pathogenic gas exhaled by patients during the incubation period can potentially spread via the ventilation system or natural ventilation, such as windows and doors. In numerous cases, indoor air quality is closely correlated with the concentration of outdoor air pollutant concentrations [10,11]. Once the air is polluted by COVID-19 viruses, there will be serious community transmission. In February 2020, health officials in Hong Kong partially evacuated residents from an apartment block, Heng Tai House, over the fears that the COVID-19 virus was being transmitted from the vent pipes of the building via the airflow in a similar manner to the SARS outbreak in Amoy Garden [12,13]. According to the preliminary analysis, the virus sample was first found near the water closets inside the bathrooms of the confirmed patients. Subsequently, the virus sample was found inside the rooftop vent pipes which are connecting to the wastewater system. Understandably, the bio-contaminants released from the waste pipes are

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<https://doi.org/10.1016/j.buildenv.2022.109215>

Received 25 February 2022; Received in revised form 8 May 2022; Accepted 17 May 2022

Available online 28 May 2022

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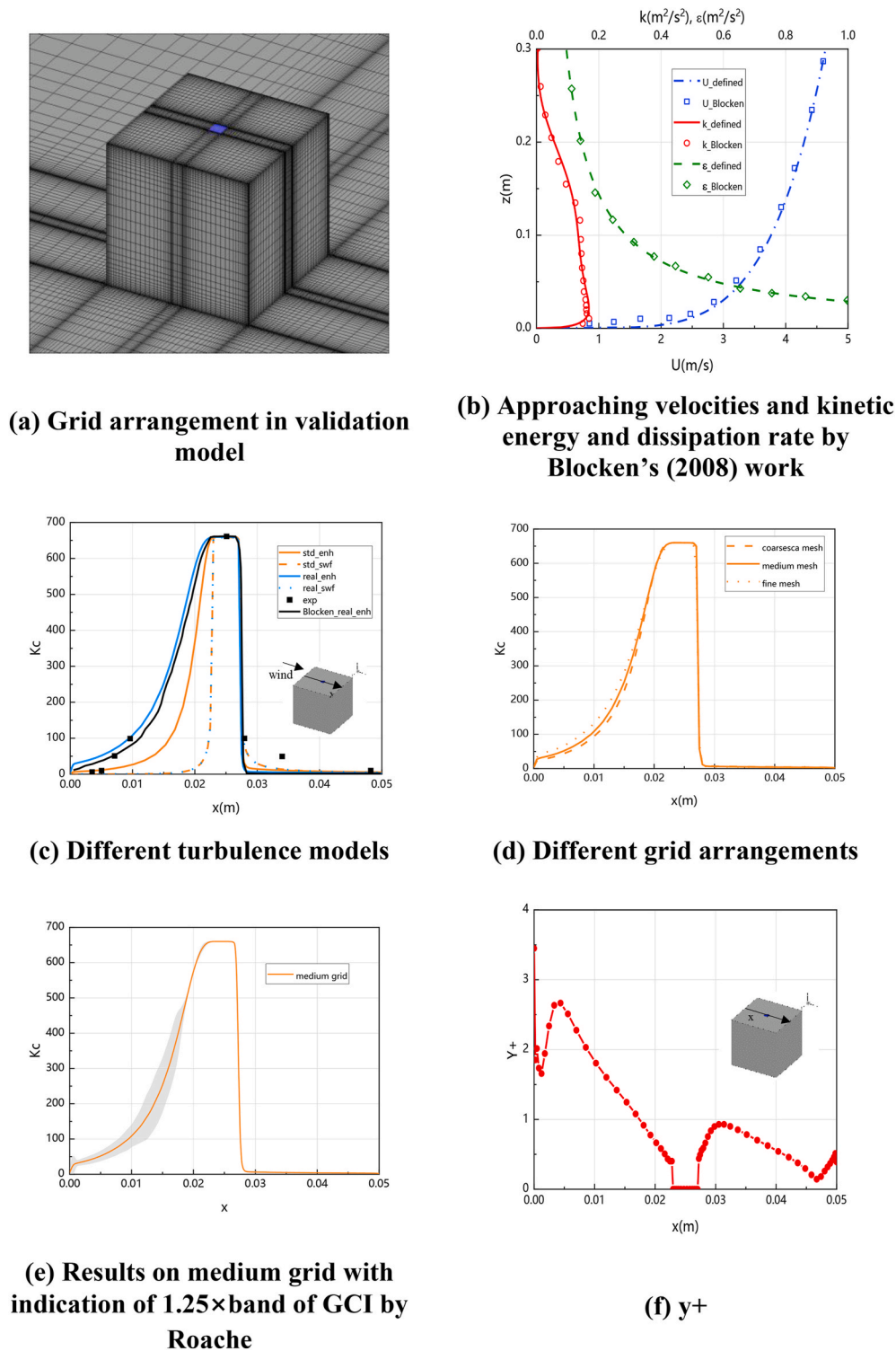


Fig. 1. Turbulence models and grid sensitivity analysis for validation model.

finally discharged from the rooftop vent pipes and dispersed to the surrounding residential area, driven by the airflow generated by wind-structure interaction. Therefore, there is an urgent need to study comprehensively the role played by wind-structure interaction in the virus transmission from the vent pipe to residential units.

1.2. Literature review

The rapid mixing of pollutants released from elevated stacks into the

wake-flow region [14], and wind tunnel experiments around rectangular buildings demonstrated the complexity of the building wake and its variability with building shape and wind direction [15]. [16] described a wind tunnel study to examine the influence of elongated rectangular buildings [17]. reported the improved estimations of the plume spreading rate, effective wind speed, turbulence intensity, and shortened projected building dimensions.

Computational fluid dynamics (CFD)-based numerical models have been increasingly used to investigate the spread of pollutants in urban

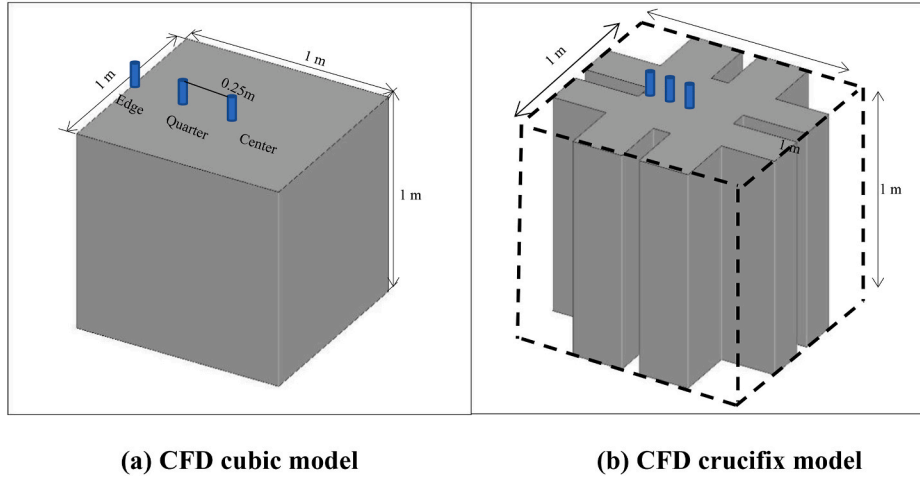


Fig. 2. Target building models with locations of vent pipes.

areas. Blocken and Carmeliet (2004) indicated that it is particularly important to select appropriate turbulence models and boundary conditions to produce accurate simulation results. Reynolds-averaged Navier–Stokes (RANS) models have been widely accepted for the simulation of pollutant dispersions in urban areas [18]. [19] discussed the influence of different turbulence models on the simulation results of pollutant dispersion with different plume buoyancies. Jeong et al. [20] investigated the effect of atmospheric stability on the near-field pollutant dispersion from the rooftop emissions of a single cubic building [21]. introduced and evaluated a new modeling approach for improving the building downwash parameterization [22,23]. found that different aspect ratios, heights, and building types have a significant impact on the spread of air pollutants in the built environment. In addition to building configurations, the location from which pollutants are discharged can also significantly impact the dispersion of pollutants.

[24] found that windward pollutant emissions migrate downward owing to the downwash created by the wind. In contrast, the dispersion of leeward pollutants is dominated by intense turbulent mixing in the near wake and is characterized by the upward migration of pollutants in the leeward direction. When the angle of wind incidence is approximately 90°, pollutants will not disperse upward or downward but remain suspended around the position of discharge.

In addition, the active botanical biofilters or functional green walls can efficiently improve indoor air quality by retaining particulate matter [25,26], such as virus droplets, and the plant-wall system helps to remove both gaseous and particulate air pollutants [27]. have researched pressure drop and air flow rate through green wall modules, and experimental work has been conducted to investigate the effect of green wall modules on air temperature and humidity.

1.3. Research gaps and objectives

Our previous work, including the studies by Refs. [22,28]; and [24]; comprehensively investigated the scenarios of air pollutants released from residential rooms. However, as aforementioned, the air pollutants released from the building roof vent pipes may play an equally important role in the dispersion of air pollutants and the spread of disease. The objective of this research was to investigate the effects of different building shapes, angles of wind incidence, locations, and heights of the roof vent pipes on the dispersion of air pollutants and the potential spread of disease.

2. Method

Before initiating any CFD work, it is vitally important to select proper

fluid models, check the grid independence, and impose reasonable boundary conditions. These CFD details have been carefully tested and validated in this study.

2.1. Validation model for roof emission

In our previous work [28], we have studied the effect of angle of wind incidence on air pollutant dispersion around a cube building and crucifix-form building, the air pollutant discharge is located on the building side. However, roof emission is quite different from side emission. To facilitate the selection of an appropriate turbulence model for the roof emission, Li and Meroney's [29] experimental model was chosen as a validation model, which was also numerical evaluated by Blocken [30]. We have created a CFD domain to match his experimental conditions, the domain is with dimensions $L \times B \times H = 1 \times 0.5 \times 0.5 \text{ m}^3$. A 0.05 m cube is located at 0.2 m from the inlet. The grid arrangement is shown in Fig. 1(a).

For the ground surface, the distance z_p is 0.0013 m . The inlet profile is a power law with exponent $\alpha = 0.19$. Mean wind speed U_H at cube roof height z_H is 3.3 m/s , z_0 is $7.5 \times 10^{-5} \text{ m}$ and u^* is 0.22 m/s . According to Li's experiment [29], the tracer gas, pure helium, was released at a discharge point with a flow rate of $12.5 \text{ cm}^3/\text{s}$ at an exit velocity ratio $U_e/U_H = 0.19$, and the discharge point was a square patch with an area of 0.1993 cm^2 .

The air pollutant concentrations at different locations were scaled based on a non-dimensional concentration:

$$K_c = \frac{CU_H H^2}{Q} \quad (1)$$

where C is the computed mean concentration, U_H is the mean wind speed at building model height H , and Q is the volumetric flow rate of the tracer gas.

A user-defined function was used to set the value of the U, k, ε in FLUENT 19.0, as shown in Fig. 1(b).

$$U(z) = U_H \times \left(\frac{z}{z_H} \right)^\alpha \quad (2)$$

$$k(z) = a(I_u(z)U(z))^2 \quad (3)$$

$$\varepsilon(z) = \frac{u^{*3}}{\kappa z} \quad (4)$$

In this function, $I_u(z)$ is a fitted streamwise turbulence intensity function:

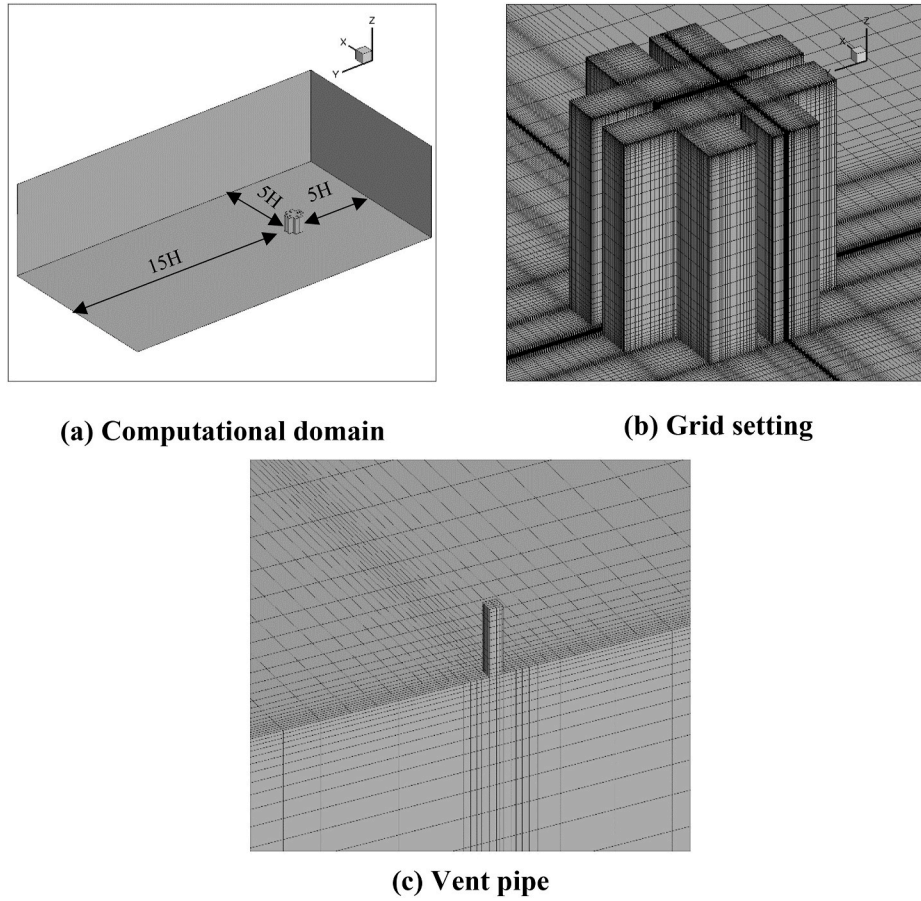


Fig. 3. Computational domain and grid arrangement for target model.

$$I_u(z) = A_0 + A_1 z + A_2 z^2 + A_3 z^3 + A_4 z^4 \quad (5)$$

where $A_0 = 0.018022$, $A_1 = -1.95298$, $A_2 = 18.41539$, $A_3 = -84.81477$, and $A_4 = 129.98937$. α is a parameter assumed to be 1.0. κ is the von Karman constant (0.39).

Different turbulence models were compared against Blocken's evaluation work [30], and the results of pollutant dispersion are compared and shown in Fig. 1(c). The results suggest that roof emission

concentrations can be reasonably modeled by the Realizable $k - \varepsilon$ model with an enhanced wall function, which is consistent with Blocken's previous report [30].

Three grids of successive refinement with the smallest cell dimension 3.3 mm, 2.0 mm, and 1.2 mm respectively, were developed to study grid independence, as shown in Fig. 1(d). For the grid convergence study, we adopted the concept of the grid-convergence index (GCI) by Roache [31, 32]. The calculated GCI is shown in Fig. 1(e), confirming that the medium-size grid (the -2mm suffix) is sufficient for the numerical simulations. The y^+ value along the cube roof is shown in Fig. 1(f). The use of these cell dimensions balances the simulation accuracy and the computational time.

2.2. CFD settings for target model

2.2.1. Building models

According to our previous work [28], a wind tunnel experiment was carried out in a boundary layer wind tunnel at the CLP Power Wind/Wave Tunnel Facility (WWTF) at Hong Kong University of Science and Technology (HKUST). The test section is 41 m in length, 4 m in height, and 5 m in wide. To minimize the requirements of blockage correction, the blockage ratio in the tunnel was maintained at approximately 5%. A generic cubic building and crucifix-form building model with the same outer envelope of 30 m in each dimension were scaled down to 1:30 for the CFD simulations, as shown in Fig. 2(a). The crucifix-form building is widely used in residential buildings in Hong Kong, as shown in Fig. 2(b). Three locations of air pollutant discharge on the roof were considered: the edge of the roof, the center of the roof, and a position lying between the edge and the center of the roof (quarter). The vent pipe was located on the top of a cubic building model, marked by a square. Three vent

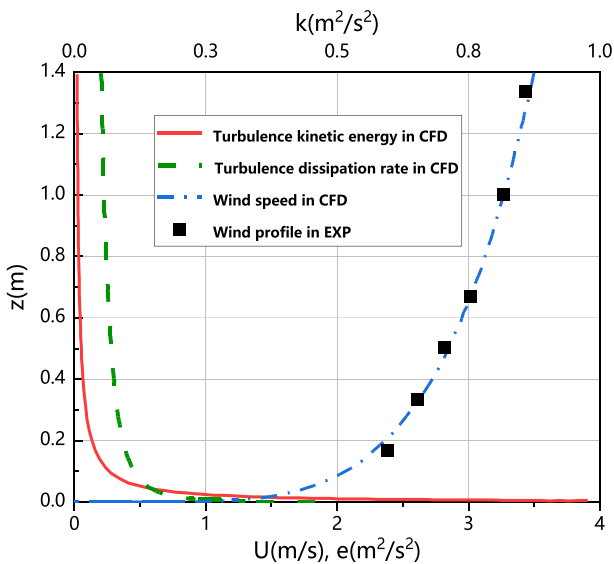


Fig. 4. Approaching velocities and kinetic energy and dissipation rate.

pipe heights in the prototype relative to the building roof were tested: 0 m, 1.5 m, and 3 m.

2.2.2. Computational domain and grid arrangement

The computational domain is shown in Fig. 3(a). The distance from the buildings to the side, inlet, and to the top of the domain is $5H$ and the distance to the outlet is $15H$, where H is the height of the building [19], in this paper H equals to 1 m.

In this study, we use our previous work's building model and grid [28], the smallest cell dimension near the edges of the buildings is $5\text{mm} \times 5\text{mm} \times 5\text{mm}$ in the x , y , and z directions, with dimensions of $2\text{mm} \times 2\text{mm} \times 2\text{mm}$ near the location at which the pollutants are released. The non-uniform and fully structured computational grid is shown in Fig. 3 (b). A maximum stretching ratio of 1.2 was applied to reduce the grid density in the areas distant from the region of interest. The locations from which air pollutants were emitted are shown in Fig. 3 (c).

2.2.3. Boundary conditions

The approaching velocities are shown in Fig. 4, which can be described by a logarithmic law (Eq. (6)), wherein u_{ABL}^* is determined based on the reference velocity at the height of the building (3.27 m/s at 1 m). According to Ref. [33] and Perén et al. [34], κ is the von Karman constant (0.42) and z is the coordinate indicating the height. The aerodynamic roughness length $z_0 = 0.66\text{ mm}$ is obtained based on curve fitting. The turbulent kinetic energy k is calculated by substituting the approaching velocities and the measured turbulence intensity into Eq. (7), where I_u is the streamwise turbulence intensity (Eq. (5)) and α is a parameter assumed to be 1.0 . The turbulence dissipation rate ε is given by Eq. (8), where C_u is an empirical constant considered as 0.09 .

$$U(z) = \frac{u_{ABL}^*}{\kappa} \ln\left(\frac{z + z_0}{z_0}\right) \quad (6)$$

$$k(z) = \alpha(I_u(z)U(z))^2 \quad (7)$$

$$\varepsilon(z) = \frac{u_{ABL}^{*3}}{k(z + z_0)} \quad (8)$$

The values of the roughness parameters, i.e., the sand-grain roughness height k_s (m) and the roughness constant C_s , are determined using their consistent relationship with the aerodynamic roughness length z_0 derived by Ref. [35]; as shown in Eq. (9).

$$k_s = \frac{9.793z_0}{C_s} \quad (9)$$

The value of k_s is restricted in FLUENT; it should be smaller than y_p , which is the distance between the center point of the wall-adjacent cell and the wall. For the ground surface surrounding the modeled buildings, k_s is 0.93 mm and consequently, the value of C_s should be 7 to satisfy Eq. (9) when $z_0 = 0.66\text{ mm}$. It should be noted that in Fluent, the required cell height can be limited to some extent by maximizing C_s but that this will often not be satisfactory [35]. A user-defined function was used to set the value of the constant C_s outside the standard allowable interval of $[0; 1]$ in FLUENT 19.0. The surface of the building was considered to have no roughness height ($k_s = 0\text{ m}$), as recommended by Perén et al. [36].

Static pressure of zero was imposed at the outlet plane. Slip wall conditions, i.e., zero normal velocity and zero normal gradients for all variables, were imposed for the top and sides of the domain.

2.2.4. Physical models of particles/droplets in the air

Virus droplets are transported via airflow in the form of droplets produced at the time of breathing/sneezing/coughing/and talking [37, 38]. The droplets with a diameter smaller than $5\text{ }\mu\text{m}$ can remain suspended in the air for a long time and facilitate distant transmission via airflow [38–40].

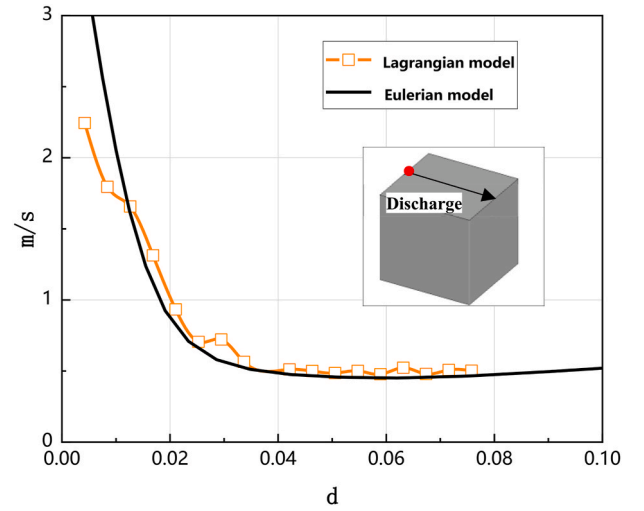


Fig. 5. Comparison of Lagrangian and Eulerian method in dispersion velocities.

Different physical models have been applied to represent virus droplets in the airflow. Some researchers treated droplets as a discrete phase and used the Lagrangian method to track the droplets' movement and dispersion [41], whereas other researchers treated droplets as a gas component and used the Eulerian method to calculate the virus droplets' dispersion [4,24,28,33]. To investigate the model differences in this study, we chose a cube model with a 0 m pipe installed at the roof edge, and air pollutant velocities were simulated using both Lagrangian and Eulerian models. The velocities at 8 mm above the cubic building roof are compared and displayed in Fig. 5, where d is the distance from the discharge point. Evidently, pollutant velocities obtained using both models are almost indistinguishable, indicating that both Lagrangian and Eulerian models can be used to represent droplets' movement in the air. This result can be explained from the perspective of multiphase fluid theory. According to the multiphase fluid theory, the particle relaxation time is

$$\tau_r = \frac{\rho_p d_p^2}{18\mu} \quad (10)$$

where ρ_p , d_p and μ are particle density, particle diameter, and gas dynamic viscosity, respectively. In the current study, the droplets' diameter is about $1\text{ }\mu\text{m}$ after rapid evaporation [39,40], density is about 1.25 kg/m^3 , and the gas-dynamic viscosity is $1.8 \times 10^{-5}\text{ Pa}\cdot\text{s}$ [5]. The resulted particle relaxation time is about $3.78 \times 10^{-9}\text{ s}$, meaning that within a very short time droplets will follow the gas flow. Thereby, the differences in velocity between the gas and the droplets can be neglected.

To simplify the simulation process and be consistent with previous studies of airborne disease spreading [4,24,28,33], the Eulerian model was applied in this study.

In the CFD model, a pollutant source is represented by a mixture of 90% air and 10% propane. A volumetric flow rate of 58.5 mL/s is placed at the top of the vent pipe to simulate the discharging process. The same trace gas was used in our previous wind tunnel and numerical studies [4, 24,28,33].

2.2.5. Solver settings

The simulations were performed using the CFD code FLUENT 19.0. The 3D steady RANS equation was solved using the Realizable $k - \varepsilon$ model. The SIMPLE algorithm was used for pressure-velocity coupling with second-order pressure interpolation. QUICK discretization schemes were utilized for the convection terms, while second-order center discretization schemes were used to describe the viscous terms in all the governing equations, including momentum, k , and ε , as well as the species transportation equations. Convergence was assumed to occur

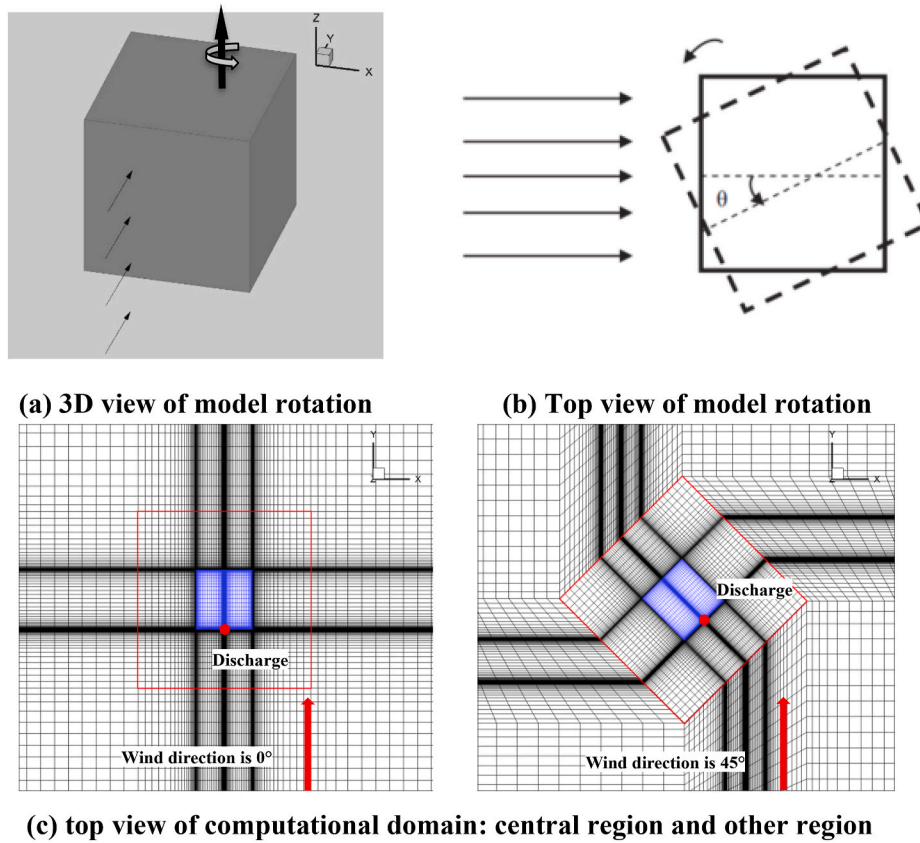


Fig. 6. Angle of wind direction.

when all the scaled residuals level off and reach a minimum of 10^{-6} for x , y , and z momentum, k , ϵ , and the species present.

2.2.6. Computational grid resolution and turbulence model choice

By comparing with Blocken's work (2008) and Li's work (1983), we validated different turbulence models for roof emission cases. The results suggest that emission concentrations can be reasonably modeled by the Realizable $k - \epsilon$ model with an enhanced wall function. For the grid-convergence index (GCI), we have confirmed that the medium-size grid (the 2 mm suffix) is sufficient for the numerical simulations in our previous work [28].

2.2.7. Rotation of building model in the CFD simulation

A wind direction of $\theta = 0^\circ$ was specified in the positive direction of the Y-axis, as shown in Fig. 6(a). The wind direction was changed by rotating the building along the Z-axis, as shown in Fig. 6(b). Each building was rotated from 0° to 180° in increments of 45° , with a total of five rotations.

To avoid interpolation of the CFD results, which is necessary when a dynamic mesh is used to represent the building model rotation, a new set of grids was generated each time the building model is rotated. To reduce the uncertainty introduced by each new grid in different wind directions, the computational domain was separated into two zones: the central region includes the building, which is $1H$ away from the side surface of the modeled building, and $4H$ from the top of the building model, as shown in Fig. 6(c), colored by red lines. The other region was considered as the rest of the domain.

3. Results

The numerical simulation results are divided into two parts: the cube model and the crucifix model. In each model, we first discuss the impact

of wind incidence angles on the air pollution dispersion, followed by the impact of air pollutant discharge location on the air pollution dispersion. Finally, we discuss the effect of increasing the vent pipe heights on air pollution dispersion. The air pollutants at the vent pipes installed at the edge, quarter, and center positions were simulated for each wind direction angle at heights of 0 m, 1.5 m, and 3 m.

For detailed simulation cases, please refer to Table 1. Where symbol " $\square$ " represents cube building models, " $+$ " symbolizes crucifix building modes, and the red dot expresses the air pollutant discharge location.

3.1. Air pollutant dispersion around a cubic building model

A cubic building model is generally used to represent generic buildings in urban areas. This simplification of the building shape yields considerable savings in terms of experimental and computational costs. Five wind incidence angles were tested in this study: 0° , 45° , 90° , 135° , and 180° .

According to Ref. [6]; even for a dilution rate of 10 000 times, the virus remnants may still be infectious in some circumstances. Since the K_c value at the discharge point is about 5500 according to Eq. (1), we have conservatively used $K_c > 0.5$, which is approximately 10 000 times diluted air pollutants, to mark a hazardous region. To further demonstrate the impact of pollutants on building dwellers, we defined Sratio. The Sratio equals the affected area colored by yellow ($K_c > 0.5$) divided by the total surface area colored by blue, as shown in Fig. 7. Apparently, the higher the Sratio, the more negative influence on the building dwellers. This is a cautious approach to include all potentially dangerous areas. The cases in which the vent pipe is at a height of 0 m are presented in Fig. 8, where the iso-surface represents $K_c = 0.5$ is colored according to the velocity ratio U/U_H , where $U_H = 3.27\text{m/s}$ is the wind speed at height H . The Sratio histogram of these cases is given in Fig. 8. The same manner of iso-surface will be used throughout this paper to identify

Table 1
Simulation cases.

Angle	Height	Position					
		Edge		Quarter		Center	
0°	1m						
	1.5m						
	3m						
45°	1m						
	1.5m						
	3m						
90°	1m						
	1.5m						
	3m						
135°	1m						
	1.5m						
	3m						
180°	1m						
	1.5m						
	3m						
Case		1, 4, 5	6, 9, 10	2	7	3	8

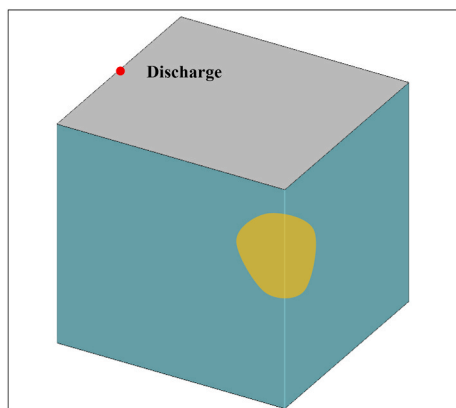
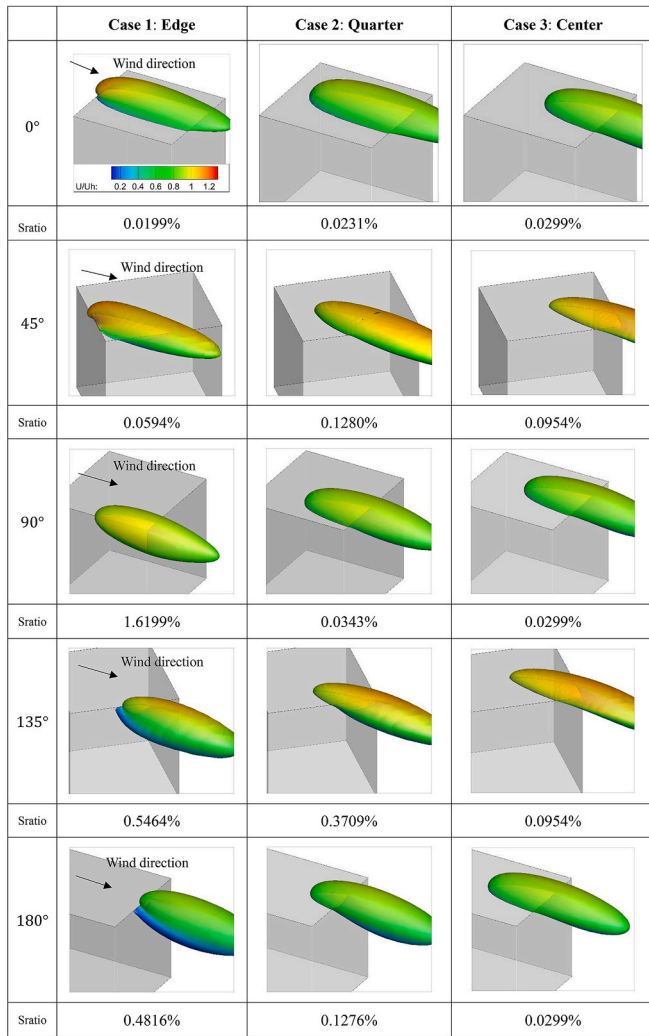


Fig. 7. Sratio equals to affected area colored by yellow divided by total surface area colored by blue. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

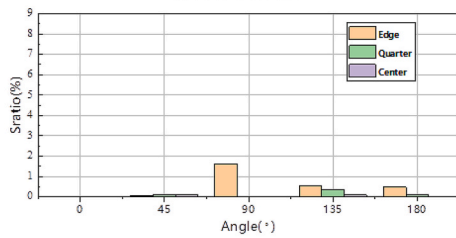
dangerous regions. **Case 1** represents the situation in which air pollutants are discharged from the roof edge; in **Case 2**, air pollutants are discharged from the quarter position, and in **Case 3**, air pollutants are discharged from the center position.

In **Case 1**, the air pollutants only affect the roof area when the wind incidence angle is 0°. As the angle of wind incidence increases, the dispersion plume progressively shifts to the corner of the roof, and the affected area gradually decreases. When the wind incidence angle reaches 90°, part of the roof flow is entrained over the roof edge onto the side of the building, and the air pollutants begin to affect the top floors closest to the roof. Air pollutants continue to be entrained and affect the higher floors at wind incidence angles of 135° and 180°.

In **Case 2**, the air pollutants also only affect the roof area when the wind incidence angle is 0°. However, the affected area is over a noticeably smaller area than that for **Case 1**. As the angle of wind incidence increases, air pollutants disperse downwind from the roof. At wind incidence angles of 90°, 135°, and 180°, the roof area continues to be affected by the dispersing air pollutants, whereas only relatively small areas of the highest floors on the side and leeward faces of the



(a) velocity contour



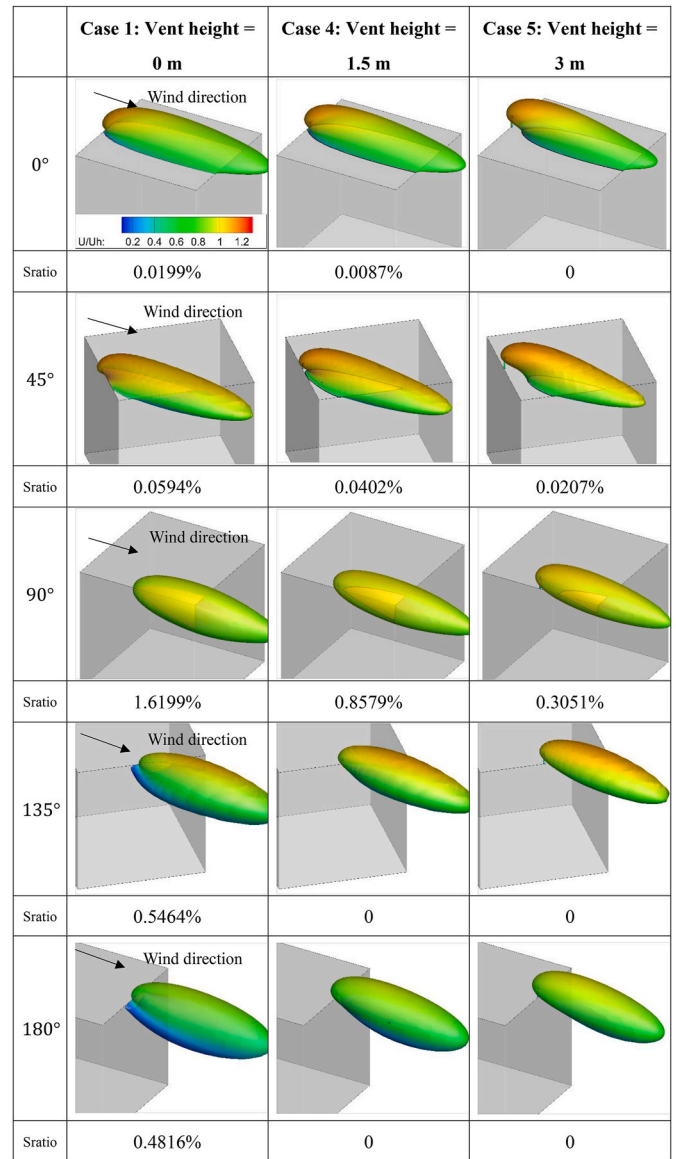
(b) Sratio

Fig. 8. Dispersion of air pollutants around cubic building model. (three installation positions and five angles of wind incidence with vent pipe installed at 0 m).

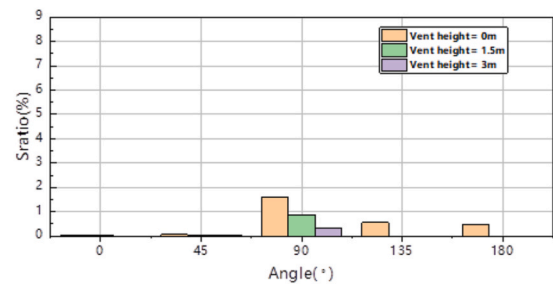
building are affected.

The pathway by which the air pollutants are dispersed in **Case 3** is like **Case 2**, with dispersed air pollutants mainly affecting the roof area. The side and leeward face of the building is only minimally affected by the dispersing air pollutants.

A comparison of **Cases 1, 2, and 3** reveals that the wind incidence angle and position of discharge play a significant role in the dispersion of air pollutants. The results suggest that the dispersing air pollutants mainly affects the roof area. Although a high air pollutant concentration on the roof may not pose a direct hazard as the roof is not generally inhabited, the re-ingestion of vent exhaust into the air-conditioning system is a potentially serious health hazard. **Case 1** is of particular importance for a wind incidence angle greater than 90° when flow



(a) velocity contour



(b) Sratio

Fig. 9. Dispersion of air pollutants around cubic building model with edge discharge. (three emission heights and five angles of wind incidence).

entrainment causes the dispersing air pollutants to migrate toward the highest floors on the side and leeward faces of the building, thus potentially impacting the occupants. Therefore, we further investigated the effects of increasing the vent pipe height from 0 m to 1.5 m and 3 m to determine whether changing the height of the vent pipe can help reduce the affected area.

Fig. 9 shows the results of air pollutant dispersion when the pollutants are discharged from the roof edge at vent heights of 0 m (**Case 1**),

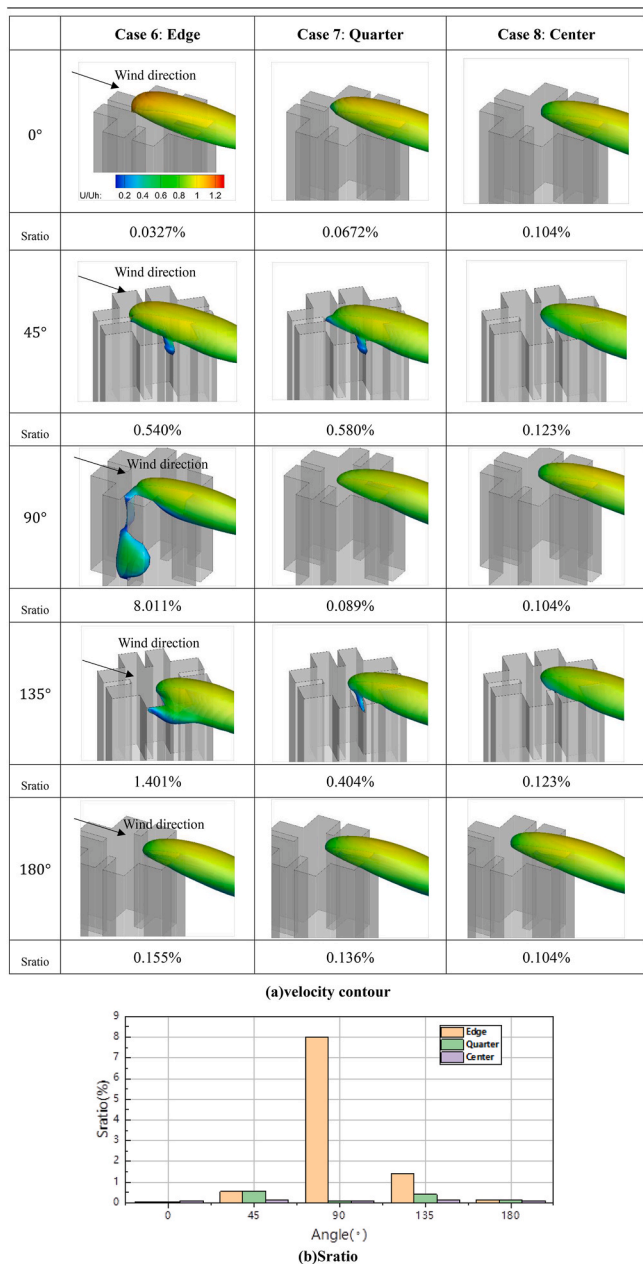


Fig. 10. Dispersion of air pollutants around crucifix-form building model. (three installation positions and five angles of wind incidence with vent pipe installed at 0 m).

1.5 m (**Case 4**), and 3 m (**Case 5**) for wind incidence angles of 0° , 45° , 90° , 135° , and 180° . At a wind incidence angle of 90° and a vent pipe height of 1.5 m, the area on the side face of the building that is affected by the dispersing air pollutant is significantly reduced compared to when the vent pipe is at 0 m. Increasing the height of the vent pipe further to 3 m can further reduce the affected area to only a small region on the side of the building. When the wind incidence angle reaches 135° and 180° , increasing the vent pipe to 1.5 m and 3 m can eliminate the area affected by air pollutants on both the side and the leeward faces of the building. The Sratio histogram of these cases is given in Fig. 9.

The air pollutant dispersion around a cubic building suggests that occupants of higher floors on the side of the building are the most vulnerable to the dispersion of air pollutants when the vent pipe is at the roof edge. Relocating the vent pipe position downstream of the windward roof can significantly reduce the affected area on the side and leeward faces of the building. Furthermore, increasing the height at

which the vent pipe is placed above the roof can substantially reduce the impact of dispersion of air pollutants. Increasing the vent pipe height to at least 1.5 m can almost eliminate the adverse influence of the dispersing air pollutants, except at a wind incidence angle of 90° , for which only a relatively small area around the highest floors on the side face of the building remains affected.

3.2. Air pollutant dispersion around a crucifix-form building model

The replica of the crucifix-form model is of the same size and configuration as the physical model tested in the wind tunnel [33]. This building model is a typical Hong Kong high-rise building. According to Hong Kong building standards, each room in a high-rise building, including bathrooms and kitchens, must have a window. To meet this standard, the test building has four vertical air ventilation channels to accommodate those windows, and they are commonly referred to as “re-entry” because air flows into/out of this space. The results of the wind tunnel experiments and comparison with the CFD simulations were reported in our previous work [28]. Like the cubic model, three air pollutant discharging positions, three vent pipe heights, and five different angles of wind incidence were tested with the crucifix-form building model.

The cases with a vent pipe height of 0 m are presented in Fig. 10, while **Case 6** presents the results when air pollutants are discharged from the roof edge. In **Case 7**, air pollutants are discharged from the quarter position, while in **Case 8** air pollutants are discharged from the center. The Sratio histogram of these cases is given in Fig. 10.

In **Case 6**, when the wind incidence angle is 0° , the air pollutant only affects the roof, which is a non-residential area. The dispersion plume progressively shifts toward the corner of the roof as the angle of wind incidence increases. When the angle of wind incidence reaches 45° , the air pollutant dispersion begins to affect the top floor units in the neighboring point of re-entry, the Sratio of the affected area is only about 0.540%. When the wind incidence angle reaches 90° , a considerable amount of air pollutants is entrained into the re-entry of the building immediately below the position of discharge and the Sratio rapidly increases to 8.011%. In this case, occupants in the high and middle floors of the re-entry of the building are exposed to the pollutant. Meanwhile, the wind blows the remaining air pollutants downstream across the edge of the roof. However, due to wind-structure interactions, the air pollutants that are dispersed downstream are mostly entrained leeward, re-entering the building re-entry, and affecting the residents in the high-floor areas on the leeward face. The area affected by the air pollutants is reduced when the angle of wind incidence reaches 135° , but pollution still engulfs the high-floor areas lying below the discharging position with the Sratio of 1.401%. When the angle of wind incidence reaches 180° , air pollutants continue to affect a sizable area leeward, where the plume re-enters the building re-entry. However, in comparison, the areas affected by the pollutants are much smaller than in **Case 1** under the same wind incidence angle in the cubic model, as shown in Fig. 7.

In **Case 7**, with the air pollutants discharged from the quarter position, air pollutants only affect the non-residential area on the roof at a wind incidence angle of 0° , with the Sratio of 0.0672%. When the wind incidence angle reaches 90° , because the air pollutant discharge position is not at the point of re-entry, the air pollutant is not entrained to re-enter the building as observed in **Case 6** but only affects the non-residential area of the roof, with the Sratio of 0.0580%. When the angle of wind incidence reaches 135° , air pollutants start to re-enter the building on the highest floors close to the discharge point. However, both the affected area and air pollutant concentration are substantially reduced compared to that in **Case 6**, with the Sratio of 0.089%, significantly smaller than 8.011% in **Case 6**. When the angle of wind incidence reaches 180° , air pollutants only affect the non-residential roof area, with the Sratio of 0.404%, significantly smaller than 1.401% in **Case 6**.

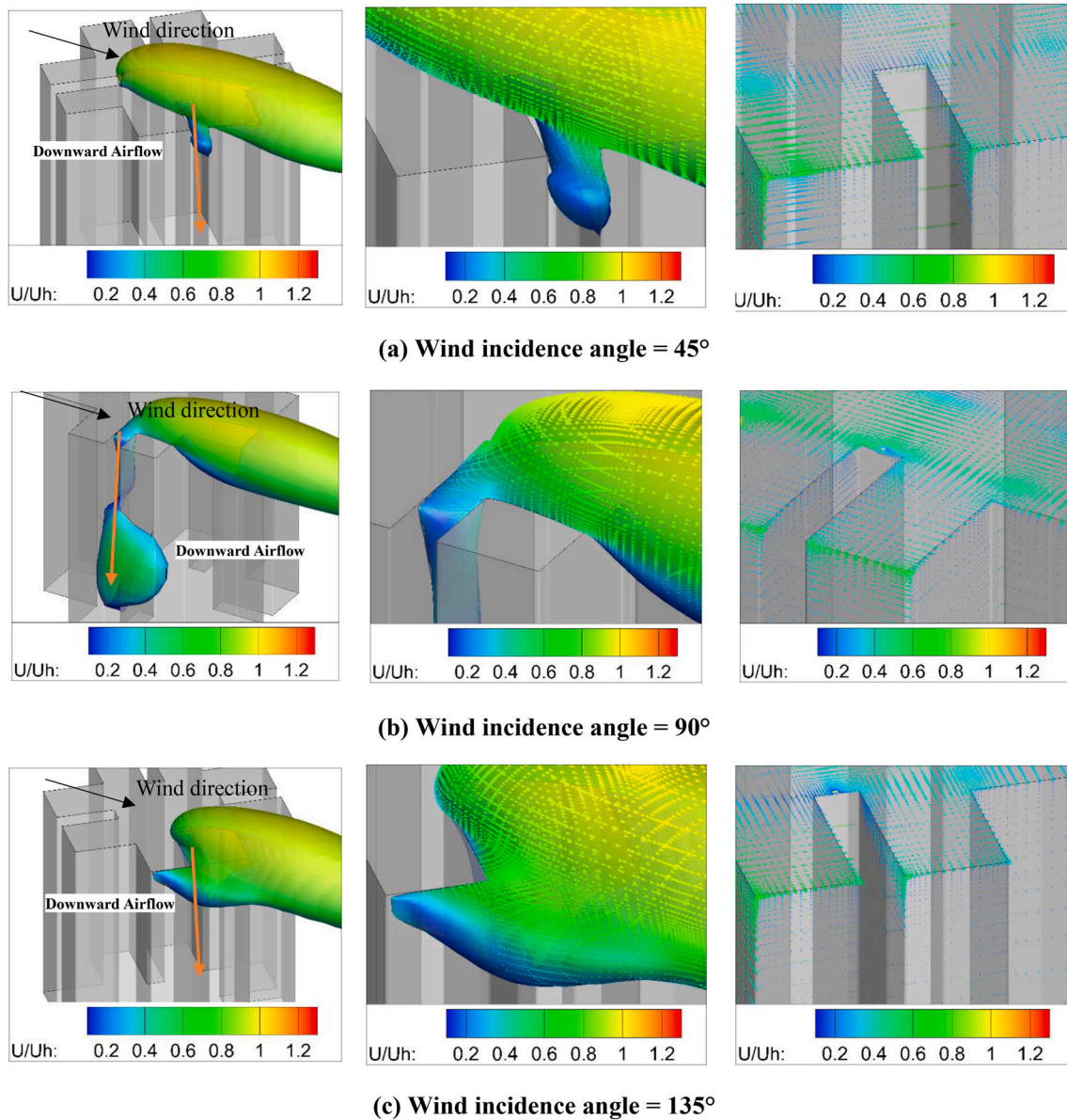


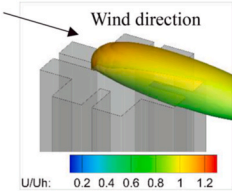
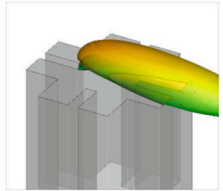
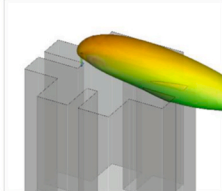
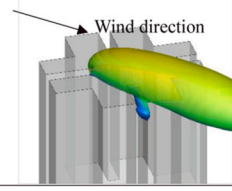
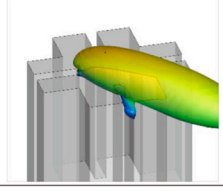
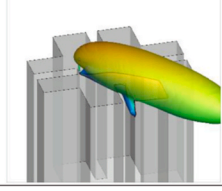
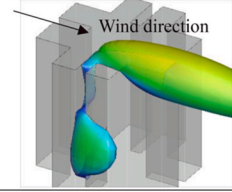
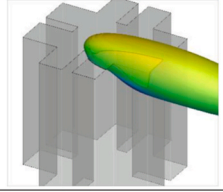
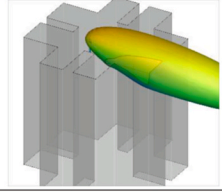
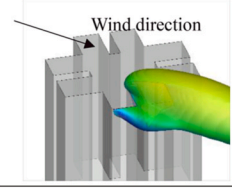
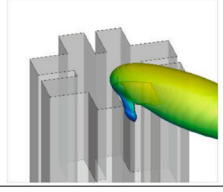
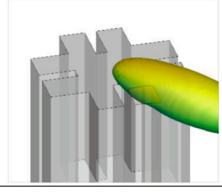
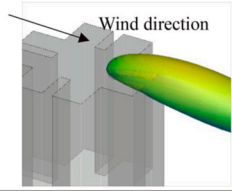
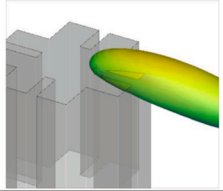
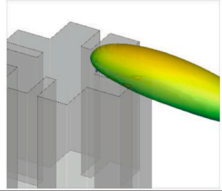
Fig. 11. Air flow and air pollutant dispersion around crucifix-form building with edge discharge (three angles of wind incidence with vent pipe installed at 0 m).

In **Case 8**, with the air pollutants discharged from the center, the migration of the dispersion plume is like that observed in **Case 7** as the angle of wind incidence increases. However, when the angle of wind incidence reaches 45° and 135°, air pollutants do not affect the high-floor area within the re-entry on the side of the building, as observed in **Case 6**; the *Sratio* is further reduced to 0.123%. Meanwhile, the high floors on the leeward face remain affected.

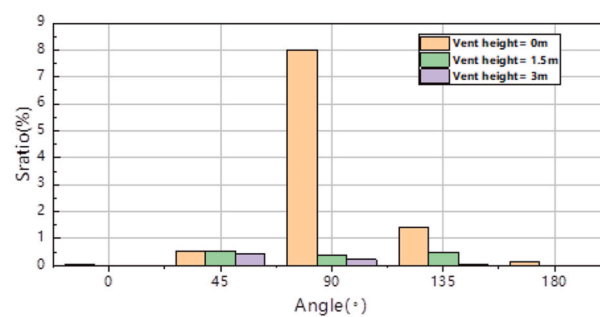
One major difference between the crucifix-form and cubic building models is the re-entry structure featured in the crucifix-form building becomes a channel that attracts a vast volume of pollutants, especially in **Case 6** at 90° angle of wind incidence. Fig. 11 further compares the wind flow pattern generated by the wind-structure interaction and the behavior of the air pollutant plume around the crucifix-form building model in **Case 6**. A downwash airflow is generated in the re-entry area that drags air pollutants into the re-entry when the angle of wind incidence reaches 45°. The downwash reaches its maximum intensity when the angle of wind incidence reaches 90°. At an angle of wind incidence of 135°, the downwash decreases, and the entrained pollutants are greatly reduced. This result reveals the dominant role of wind-structure interactions in the dispersion of air pollutants.

A comparison of **Cases 6, 7, and 8** reveals that both the wind incidence angle and the position of discharge play a significant role in the dispersion of air pollutants in the crucifix-form building model. The results also suggest that high air pollutant concentrations within the re-entry pose a potentially serious health hazard for the occupants in the middle and high floors units. The worst scenario is **Case 6**, wherein wind-structure interaction causes a significant amount of the airflow to migrate into the building re-entry, entraining air pollutants towards the high and middle floors. Evidently, relocating the position of discharge can reduce the re-entry area affected by entrained air pollutants, but this occasionally leads to an increase in the affected area on the leeward face.

Like the cubic building model, we also tested cases in which the height of the vent pipe is increased on the crucifix-form building model. Fig. 12 shows the comparison of the results for conditions under which air pollutants are discharged from the roof edge at vent heights of 0 m (**Case 6**), 1.5 m (**Case 9**), and 3 m (**Case 10**). **Case 9**. It demonstrates that *Sratio*, which represents the area proportion affected by air pollutants, is significantly reduced. The *Sratio* histogram of these cases is given in Fig. 12 And the affected area both on the roof of the building and within the re-entry structure is reduced when the vent pipe is at a

	Case 6: Vent height = 0 m	Case 9: Vent height = 1.5 m	Case 10: Vent height = 3 m
0°			
Sratio	0.033%	0.0130%	0
45°			
Sratio	0.541%	0.511%	0.411%
90°			
Sratio	8.011%	0.396%	0.221%
135°			
Sratio	1.401%	0.501%	0.032%
180°			
Sratio	0.155%	0.012%	0

(a) velocity contour



(b) Sratio

Fig. 12. Dispersion of air pollutants around crucifix-form building model with edge discharge. (three emission heights and five angles of wind incidence).

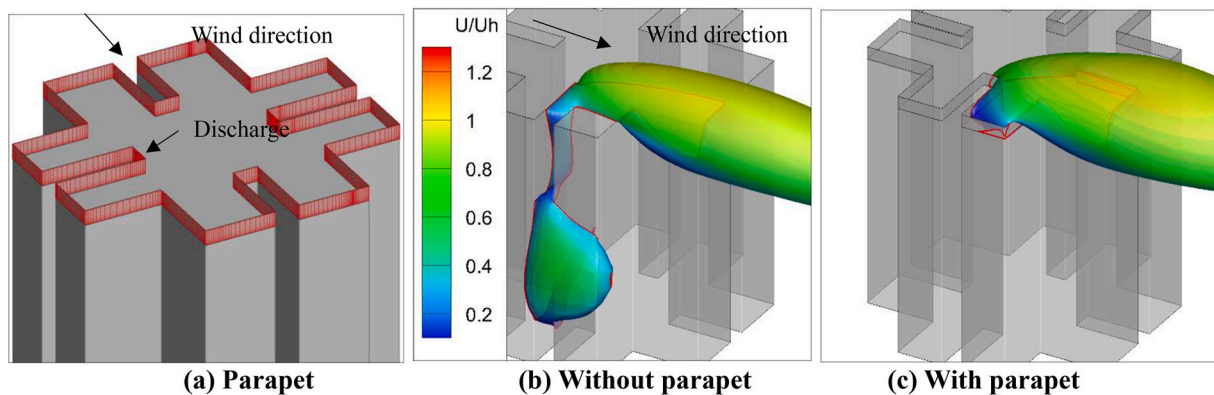


Fig. 13. Parapet impact on dispersion of roof edge discharged air pollutants.

height of 1.5 m. In particular, no pollutants migrate into the building re-entry when the angle of wind incidence reaches 90° . At wind incidence angles of 45° and 135° , air pollutants were found to migrate into the building re-entry; however, the affected areas are greatly reduced. Further increasing the vent pipe height to 3 m greatly reduces the air pollutant affected area on the roof and within the re-entry structure.

4. Discussion

As mentioned in the introduction, airborne diseases spreading from the building roof vent pipe is of great importance to investigate. Furthermore, the air pollutants dispersion around a crucifix-form building model is of particular interest, and selected important findings are further analyzed and discussed hereunder.

Although increasing the vent pipe height can effectively reduce the adverse effect of air pollutants, the vent pipe height is restricted by other factors, such as structural stability. Furthermore, Fig. 12 clearly shows that merely increasing vent pipe height cannot make the K_c at the 9th floor close to zero for all the possible angles of wind incidence. To further reduce the impact of air pollutants, additional structures that divert the airflow from the residential area are essential. We choose Case 6 for a wind incidence angle of 90° as an example. A parapet with a height of 1.5 m is installed at the roof edge, as shown in Fig. 13(a). Simulation results confirm that an additional parapet can eliminate the air pollutant in the re-entry, as shown in Fig. 13(b) (c). By introducing a parapet, the S_{ratio} in the affected area dropped sharply from 8.011% to 0.711%. The functionality of parapets or other structures in controlling air pollutant dispersion deserves future studies.

5. Limitations

Although encouraging results have been obtained, some limitations are highlighted here. First, all the presented CFD results were obtained based on wind tunnel scale simulations and only cubic and crucifix-form buildings were investigated. In future work, additional buildings with different prototype shapes and sizes will be included in CFD models for generating comprehensive and realistic pictures of the effects of wind-structure interaction on air pollutant dispersion. Second, the simulation conditions were isothermal. In future work, a range of atmospheric conditions will be included in CFD models to test the influence of meteorological conditions on air pollutant dispersion in the built environment. Third, CFD can be expensive. To estimate the air pollutant dispersion in real-life conditions, an empirical formulation that stemmed from CFD, wind tunnel experiments, and field measurement will be developed for practical applications.

6. Conclusion

CFD simulations were performed to study the air pollutant dispersion

around cubic and crucifix-form building models. Air pollutant dispersion is found to depend on the building configuration, angles of wind incidence, and the height and location of the roof vent pipe. The residential areas on the side of the building on higher floors are most susceptible to the dispersing air pollutants when the vent pipe is at the roof edge in both cubic and crucifix-form buildings. Remarkably, air pollutants were found to migrate into the building re-entry in the crucifix-form building model under certain wind incidence angles. Relocating the vent pipe position downstream and increasing the height of the vent pipe above the roof can substantially reduce the impact of the dispersing air pollutants.

Although the leading role played by the vent pipe height is confirmed, the adjustment of the vent pipe height is restricted and thus additional parapets are suggested to be installed along the building roof edges to further reduce the impact of the dispersing air pollutants. Promising preliminary results have been presented in this paper, which warrants a detailed study in future work.

The outcomes of the current study can help to control the risk of airborne contaminants/diseases spreading in current and future built environments.

CRediT authorship contribution statement

Guang Ze Guo: Writing – original draft, Validation, Formal analysis, Investigation. **Yong Yu:** Writing – review & editing, Investigation, Formal analysis. **K.C.S. Kwok:** Conceptualization, Writing – review & editing. **Yu Zhang:** Writing – review & editing, Writing – original draft, Resources, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research is supported by a special fund of the State Key Joint Laboratory of Environment Simulation and Pollution Control (2015)-15K09ESPCT, Tsinghua University, China.

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